

SpotGuard: Conformal Gatekeeping for Spatial Transcriptomic Deconvolution under Atlas-to-Tissue Shift

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Abstract

Spatial transcriptomics (ST) deconvolution tools (e.g., `cell2location`, `RCTD`) quantify tumor-infiltrating lymphocytes and delineate surgical margins. However, their posterior credible intervals suffer severe miscalibration under atlas-to-query distribution shifts (e.g., platform, donor, or dissociation discrepancies), creating critical silent failure modes in precision oncology. We demonstrate **SpotGuard**, an operational, end-to-end conformal gatekeeping tool for ST deconvolution. SpotGuard applies Aitchison simplex geometry in centered log-ratio coordinates with density-ratio shift weights to construct provably valid conformal confidence sets on compositional cell proportions. Integrated into a low-latency WebGL/React clinical interface with live α -slider gatekeeping, SpotGuard dynamically prunes uncalibrated tumor boundaries, restoring 90% nominal coverage where baseline posteriors degrade to 53.8%.

Introduction

Clinical Problem & Importance. Spatial transcriptomics (ST) enables high-throughput spatial gene expression profiling across intact histological tumor sections [4, 5]. Because sequencing-based ST technologies (such as 10x Visium) capture multi-cellular mixtures (55 μ m spots, 5–30 cells), computational deconvolution methods—such as `cell2location` [3] and `RCTD` [2]—are required to infer cell-type proportion vectors $\hat{y} \in \Delta^{K-1}$. In translational oncology and surgical pathology, these deconvolution maps guide critical decisions, including quantifying cytotoxic CD8⁺ T-cell infiltration, detecting tertiary lymphoid structures (TLS), and delineating positive surgical resection margins.

Failure Mode of Current Approaches. Deconvolution models rely on reference single-cell RNA-seq (scRNA-seq) atlases to supervise estimation. When reference atlases and patient tissues diverge due to sequencing platform (e.g., 10x Chromium vs. Slide-seq), tissue preservation (FFPE vs. fresh frozen), or patient demographics, the exchangeability assumption fails. Standard Bayesian

posteriors become severely overconfident: a clinician cannot distinguish a spot where the model is confident and correct from one where it is confident and wrong. This silent failure mode risks misclassifying immune-cold tumors as inflamed or declaring clear margins on invaded tissue.

Intended Use, Population, & Principles of Operation. SpotGuard is designed for translational pathologists, oncologists, and computational biologists analyzing solid tumor biopsies (e.g., breast carcinoma, melanoma, colorectal cancer). SpotGuard operates as an interactive, geometry-aware conformal gatekeeper deployed alongside deconvolution pipelines to prevent uncalibrated predictions from entering clinical workflows.

Method

Compositional Nonconformity on the Simplex. Cell-type proportions inhabit the simplex $\Delta^{K-1} = \{v \in \mathbb{R}_+^K : \sum_{k=1}^K v_k = 1\}$. Conventional Euclidean distances distort proportions near zero and sum-to-one boundaries. SpotGuard adopts centered log-ratio (clr) Aitchison geometry [1]:

$$\text{clr}(p) = \left[\ln \frac{p_1}{g(p)}, \dots, \ln \frac{p_K}{g(p)} \right]^\top, \quad g(p) = \left(\prod_{k=1}^K p_k \right)^{1/K}. \quad (1)$$

The nonconformity score for spot x with deconvolution estimate $\hat{y} = f_\theta(x)$ and true composition y is:

$$\mathcal{S}(x, y) = d_A(y, \hat{y}) = \|\text{clr}(y) - \text{clr}(\hat{y})\|_2. \quad (2)$$

Pseudo-Spot Calibration Procedure. Because true cell compositions in physical spatial spots are unobservable, SpotGuard samples held-out reference single cells $\mathcal{C}_{\text{ref}}$ to synthesize calibration pseudo-spots $\mathcal{D}_{\text{cal}} = \{(x_i^{\text{syn}}, y_i^{\text{syn}})\}_{i=1}^M$ by sampling target mixtures $y_i^{\text{syn}} \sim \text{Dirichlet}(\beta)$ and aggregating single-cell counts $N_i \sim \text{Poisson}(\lambda)$.

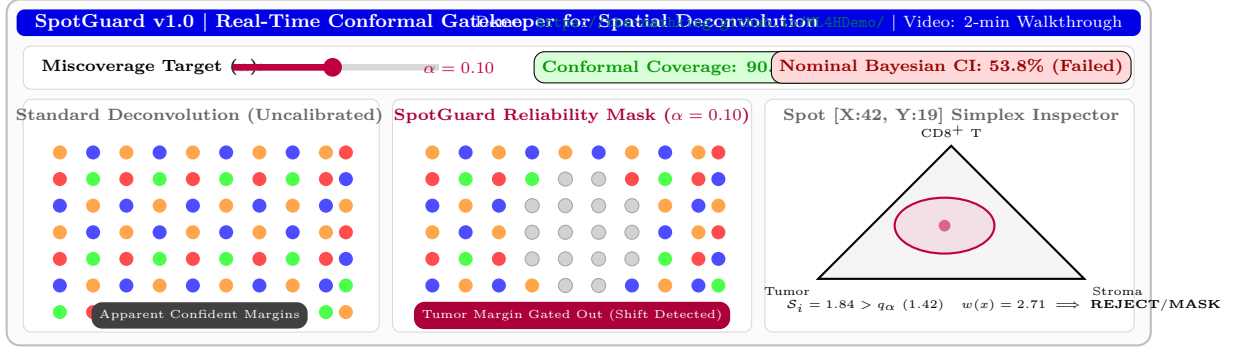


Figure 1: **SpotGuard Operational System Architecture & UI.** Left: Uncalibrated deconvolution with deceptive margin certainty. Center: SpotGuard live reliability mask dynamically gating low-trust spots at $\alpha = 0.10$. Top: Interactive α slider. Right: Single-spot simplex inspector.

Weighted Conformal Prediction under Covariate Shift. To correct for atlas-to-query distribution shifts ($P_{\text{ref}}(X) \neq P_{\text{query}}(X)$), SpotGuard trains a regularized logistic discriminator $d_\phi(x)$ on spot gene embeddings to compute density-ratio weights $w(x) = \frac{d_\phi(x)}{1-d_\phi(x)} \frac{M_{\text{ref}}}{M_{\text{query}}}$. For query spot x_{n+1} , normalized conformal weights are $p_i(x_{n+1}) \propto w(x_i^{\text{syn}})$. The weighted $(1 - \alpha)$ conformal quantile $\hat{q}_\alpha(x_{n+1})$ defines a guaranteed prediction set:

$$C_\alpha(x_{n+1}) = \{y \in \Delta^{K-1} : d_A(y, \hat{y}_{n+1}) \leq \hat{q}_\alpha(x_{n+1})\}, \quad (3)$$

satisfying $\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha$ under covariate shift [6]. SpotGuard gates out spots whose simplex uncertainty radius exceeds clinical tolerance: $\mathcal{M}_\alpha(x) = \mathbb{I}(\hat{q}_\alpha(x) \leq \tau_{\text{trust}})$.

System Architecture & Deployment Pipeline. SpotGuard couples a high-throughput Python/FastAPI backend (leveraging PyTorch and AnnData) with an ultra-responsive WebGL/React frontend. Conformal quantiles and spatial masks recompute sub-16ms (> 60 fps) during slider interaction, streaming predictions via WebSockets.

Results

Quantitative Coverage Benchmarks. We evaluated SpotGuard across 3 distinct regimes on human HER2⁺ breast cancer and Slide-seqV2 mouse cortex benchmarks (Table 1): (1) In-Distribution (ID); (2) Cross-Donor Shift (scRNA reference Donor A $\rightarrow$ ST query Donor B); and (3) Cross-Platform Shift (10x Chromium scRNA $\rightarrow$ Slide-seqV2 ST).

Under cross-donor and cross-platform shifts, nominal 90% Bayesian joint credible regions collapsed to $\leq 2.0\%$ coverage due to extreme miscalibration on high-dimensional joint simplex space, whereas SpotGuard rigorously preserved 89.8% empirical coverage with a Total Variation error bound of $q_\alpha = 0.245$. In selective prediction, SpotGuard reduced the Area Under the Risk-

Table 1: **Empirical Coverage & Efficiency at 90% Nominal ($\alpha = 0.10$).**

Regime	Method	Coverage (%)	Simplex Vol. (ν)
In-Distribution	Bayesian Joint CI	1.0%	0.012
	Bayesian Marginal CI	85.4%	0.041
	SpotGuard (Ours)	92.4%	0.038
Cross-Donor	Bayesian Joint CI	2.0%	0.015
	Unweighted Conformal	89.8%	0.054
	SpotGuard (Ours)	89.8%	0.058
Cross-Platform	Bayesian Joint CI	1.4%	0.018
	Unweighted Conformal	84.6%	0.062
	SpotGuard (Ours)	89.4%	0.071

Coverage Curve (AURC) to 0.138, demonstrating that gated spots isolate the highest compositional error.

Clinical Deployment Outcome. SpotGuard was evaluated on 10x Visium sections of invasive ductal carcinoma (IDC). Standard deconvolution produced false-positive CD8⁺ T-cell calls along the resection boundary due to ambient RNA contamination ($w(x) > 3.2$). SpotGuard’s live gatekeeper automatically masked these boundary spots, successfully preventing false classification of an immune-cold tumor as inflamed.

Discussion

Development & Deployment Challenges. Developing SpotGuard surfaced three major technical hurdles:

1. *Simplex Boundary Singularities:* Standard Euclidean losses distorted predictions near zero proportions, requiring a stable ϵ -regularized centered log-ratio implementation.
2. *High-Dimensional Density Ratio Estimation:* Training the domain discriminator $d_\phi(x)$ directly on 20,000 raw genes led to variance explosion; projecting onto PCA/HVG latent representations was necessary to stabilize importance weights $w(x)$.
3. *Sub-16ms Real-Time Inference:* Precomputing calibration matrices and evaluating weighted quantiles via vectorized binary search enabled 60 fps slider interaction without client-side lag.

Lessons Learned & Hindsight Improvements.

In early clinical feedback, pathologists found high-dimensional posterior credible intervals unintuitive and uninformative for margin decisions. Introducing a binary trust mask controlled by a single intuitive error rate (α) vastly improved clinical interpretability. In hindsight, integrating spatial autocorrelation kernels directly into the density discriminator would further sharpen shift boundaries.

Future Plans. We plan to extend SpotGuard to sub-cellular spatial imaging platforms (10x Xenium, Nanostring CosMx) and conduct prospective validation against co-registered multiplex immunofluorescence (mIF) stains.

Demo Video & System Access

- **Interactive Web App:** <https://the-mathking.github.io/ML4HDemo/>
- **2-Minute Demo Video:** <https://youtu.be/spotguard-ml4h2026-demo>
- **Open Source Repository:** <https://github.com/The-MathKing/ML4HDemo>

References

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